Verifying SimCLR: A Critical Analysis of Contrastive Learning Through Systematic Ablation

TEAM25

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Abstract

SimCLR [1] showed that contrastive learning, a self-supervised approach that learns representations by comparing different views of the same image, can produce strong visual features using three key components: strong augmentation composition, a nonlinear projection head, and large batch sizes. While these design choices are widely adopted, their robustness under limited data and compute remains unclear, since SimCLR's original results were obtained on large-scale ImageNet experiments.

In this work, we reproduce SimCLR on CIFAR-10 and STL-10 and conduct controlled ablation experiments to study four questions: (1) the importance of augmentation composition, (2) whether the projection head consistently improves downstream representations, (3) the necessity of large batch sizes compared to momentum-based alternatives, and (4) sensitivity to the temperature parameter in the contrastive loss. By isolating these components, our goal is to identify which SimCLR design principles remain robust and which are dependent on large-scale training conditions.

1 Overview and Motivation

Self-supervised learning has become a dominant approach for representation learning in computer vision, enabling models to learn useful features from unlabeled data. Among these methods, contrastive learning has emerged as one of the most effective strategies for learning visual representations.

SimCLR showed that strong representations can be learned using a relatively simple framework built around three main components: strong augmentation composition, a nonlinear projection head, and large batch sizes that provide many negative samples during training. These design choices have been widely adopted and have influenced many subsequent self-supervised learning methods.

However, SimCLR's conclusions were primarily derived from experiments on the large-scale ImageNet with substantial computational resources, including batch sizes up to 8192. In smaller-scale settings such as CIFAR-10 and STL-10, the available data, image resolution, and feasible batch sizes differ significantly. As a result, it remains unclear whether these design principles remain equally important when contrastive learning is applied under more constrained data and compute conditions.

In this project, we evaluate the robustness of SimCLR's core design decisions under small-scale conditions. Specifically, we ask:

- How critical is augmentation composition for representation learning on small datasets?
- Does the nonlinear projection head always improve downstream performance?
- How sensitive is performance to batch size and temperature?

Our goal is to achieve at least **85%** linear evaluation accuracy on CIFAR-10 under full training, consistent with previously reported SimCLR results, and then measure how controlled ablations of key components affect performance relative to this baseline.

2 Related Work

SimCLR [1] demonstrated that strong visual representations can be learned through contrastive learning by combining heavy augmentation composition, a nonlinear projection head, temperature-scaled contrastive loss, and large batch sizes. Their results showed that these components together produce strong downstream performance, but the experiments were conducted primarily at ImageNet scale using substantial computational resources.

Subsequent work has explored alternatives and theoretical perspectives on contrastive learning. MoCo

[2] introduced a momentum encoder and a queue of negative samples, allowing models to learn effective representations without requiring extremely large batch sizes. Meanwhile, Wang and Isola [3] analyzed contrastive learning through the alignment–uniformity framework, providing a geometric interpretation of how representations cluster and spread in embedding space.

While these works establish the importance of SimCLR's design choices and provide theoretical insights into contrastive objectives, the ablations have been performed on large-scale datasets. Our work builds on these studies by evaluating SimCLR's components under smaller datasets and constrained compute settings.

3 Methodology

We follow the SimCLR framework [1]:

3.1 Model Description

Component	Details
Encoder $f(\cdot)$	ResNet-18 conv1: 3×3, stride 1 (CIFAR stem) Output representation $h \in \mathbb{R}^{512}$
Projection Head $g(\cdot)$	2-layer MLP: 512 $\rightarrow$ 512 (ReLU) $\rightarrow$ 128 L2-normalized embedding z
Loss	NT-Xent contrastive loss, $\tau = 0.07$
Optimizer	LARS, lr $= 0.3 \times$ batch/256 weight decay 10^{-6}, cosine LR + 10-epoch warmup
Augmentations	RandomResizedCrop, HorizontalFlip, ColorJitter, RandomGrayscale ($p = 0.2$), GaussianBlur ($p = 0.3$)
Linear Evaluation	Frozen encoder + single FC layer SGD (momentum 0.9), cosine LR, 100 epochs

Table 1: Model and evaluation setup.

Implementation was developed in PyTorch and extended from an open-source SimCLR implementation.[1] Our modified codebase, including all experiment configurations used in this study, is publicly available at `https://github.com/SAlaMusa/IDL`.

[1] `https://github.com/sthalles/SimCLR`

3.2 Datasets

We evaluate our experiments on CIFAR-10 and STL-10. CIFAR-10 contains 50000 training images and 10000 test images across 10 classes with resolution 32×32. STL-10 contains 100000 unlabeled images used for self-supervised pretraining and 13000 labeled images used for downstream evaluation. STL-10 images have resolution 96×96, allowing us to test whether trends observed on CIFAR-10 generalize to a higher-resolution dataset.

3.3 Training Objective

Training uses the NT-Xent (Normalized Temperature-scaled Cross Entropy) loss introduced in SimCLR. For each image, two augmented views are generated and treated as a positive pair. The model learns representations by maximizing agreement between positive pairs while contrasting them with other images in the batch. Similarity between embeddings is computed using cosine similarity after L2 normalization.

3.4 Evaluation Metric

Representation quality is evaluated using the standard linear evaluation protocol. After self-supervised pretraining, the encoder parameters are frozen, and a single linear classifier is trained on labeled data. The classifier is trained for 100 epochs using SGD with momentum 0.9 and a cosine learning rate schedule. Performance is reported as top-1 classification accuracy on the test set. This metric measures how well the learned representation supports downstream classification without additional fine-tuning.

3.5 Experimental Depth and Iteration

To balance experimental coverage with computational constraints, we adopt a two-stage experimental protocol: the screening stage is designed to efficiently explore multiple experimental configurations, while the confirmatory stage validates the most informative findings under longer training schedules.

Stage	Epochs	Seeds	Purpose
Screening	200	1	Explore augmentation configurations
Confirmatory (planned)	800	3	Validate selected findings

Table 2: Two-stage experimental protocol

Initial screening experiments have already been conducted across multiple configurations. These include reproduction of the SimCLR baseline on both CIFAR-10 and STL-10 as well as augmentation ablations designed to isolate the contribution of individual augmentations:

- **Baseline reproduction:** Standard SimCLR training on CIFAR-10 and STL-10 using the full augmentation pipeline.
- **Single-augmentation ablations:** Experiments that remove individual augmentations from the pipeline to measure their contribution.
- **Augmentation interaction experiments:** Pairwise augmentation combinations and intentionally harmful augmentations designed to test the sensitivity of learned representations to view generation strategies.

4 Baseline Selection and Extensions

4.1 Baselines

Our primary baseline is the SimCLR framework [1]. The model is trained using the full SimCLR augmentation pipeline and evaluated using the standard linear evaluation protocol.

For the midterm stage, we reproduce SimCLR using a reduced training schedule (200 epochs) to obtain initial screening results. Baseline experiments have been implemented and executed on both CIFAR-10 and STL-10. Representation quality is measured using top-1 linear evaluation accuracy after freezing the encoder and training a linear classifier for 100 epochs.

These baseline runs establish a reference point for all subsequent ablation experiments. Confirmatory baseline runs using the full 800-epoch training schedule are planned for the final stage of the project.

4.2 Planned Baseline Comparisons

To add context our results, we also consider several comparison baselines commonly used in contrastive learning studies:

- Small-batch SimCLR variants to study batch-size sensitivity.
- MoCo v2 [2], which replaces large in-batch negatives with a momentum encoder and queue.
- Supervised pretraining, used as a reference upper bound for representation quality.

4.3 Implemented Extensions

Beyond reproducing the SimCLR baseline, we implement several experimental extensions to isolate which components of the framework affect performance in small-scale settings.

We evaluate:

- Removal of individual augmentations from the standard pipeline.
- Pairwise augmentation combinations to study interaction effects.
- Intentionally harmful augmentations like strong rotations, noise, solarization to test representation sensitivity.

These experiments provide a way to analyze how view generation influences contrastive representation learning. Additional experiments examining projection head depth and batch size–temperature interactions are planned for the next stage.

5 Results

5.1 Baseline Results

Dataset	Top-1 Accuracy
CIFAR-10 (200 epochs)	76.36%
STL-10 (100 epochs)	68.03%

Table 3: Baseline SimCLR results.

These results establish the reference performance used for all subsequent ablation experiments.

5.2 Single-Augmentation Ablations

To examine the contribution of individual augmentations in the SimCLR pipeline, we remove one augmentation at a time while keeping all other training settings fixed.

Configuration	Top-1 Accuracy
Baseline (full pipeline)	76.36%
No crop	47.70%
No horizontal flip	74.42%
No color jitter	68.23%
No grayscale	69.79%
No blur	74.56%

Table 4: Single-augmentation ablation results on CIFAR-10.

Removing random cropping results in the largest reduction in performance, while removing horizontal flipping or Gaussian blur produces relatively small changes. Removing color jitter or grayscale results in moderate decreases relative to the baseline.

5.3 Augmentation Interaction Experiments

We next evaluate selected pairs of augmentations to examine how different transformations interact during contrastive training.

Table 5: Pairwise augmentation experiments on CIFAR-10.

Configuration	**Top-1 Accuracy**
Crop + jitter	74.22%
Crop + grayscale	72.33%
Crop + blur	59.03%
Jitter + grayscale	41.87%

Performance varies substantially across augmentation pairs. Configurations that include cropping remain relatively strong, while the jitter and grayscale combination produces the lowest accuracy among the tested pairs.

5.4 Harmful Augmentation Experiments

To further test sensitivity to view design, we introduce augmentations that may distort semantic information.

Configuration	**Top-1 Accuracy**
Rotation (180°)	65.53%
Noise	62.42%
Solarize	78.31%

Table 6: Results for intentionally harmful augmentations on CIFAR-10.

Both rotation and additive noise reduce performance relative to the baseline, while solarization slightly improves the measured linear evaluation accuracy.

6 Discussion

The screening experiments provide initial insight into how SimCLR design choices behave in small-scale settings such as CIFAR-10. Among the tested augmentations, removing random cropping produces the largest drop in performance, indicating that spatially distinct views play an important role in the contrastive learning objective. Other augmentations, including horizontal flipping and Gaussian blur, appear to have a smaller influence on representation quality in our current setup, while color-based transformations such as jitter and grayscale contribute moderate improvements.

The pairwise augmentation experiments further suggest that augmentation interactions can significantly affect training outcomes. Some combinations remain relatively strong, particularly those that include cropping, while others lead to substantial performance degradation. These results indicate that the effectiveness of contrastive learning is sensitive not only to individual augmentations but also to how they are combined when generating positive pairs.

The harmful augmentation experiments highlight additional sensitivity in the learning process. Transformations such as large rotations and additive noise reduce performance relative to the baseline, suggesting that some view transformations may distort semantic structure in ways that make the contrastive task less informative. At the same time, the solarization experiment slightly improves performance, indicating that unconventional augmentations may still provide useful variability in certain cases.

Several challenges and risks remain at this stage of the project:

1. The current results are based on reduced training schedules used for screening, which may not fully reflect the behavior of SimCLR under longer training regimes. Previous work has shown that contrastive learning performance improves substantially with longer training and larger batch sizes [1].

2. Contrastive learning methods are known to be sensitive to implementation details, including augmentation parameters, optimizer configuration, and temperature settings [1, 3]

3. The relatively small scale of CIFAR-10 and STL-10 may limit the generality of conclusions compared to large-scale datasets, ImageNet, used in the original SimCLR study.

7 Future Directions

To strengthen the conclusions of this study, several follow-up experiments are planned:

1. We will run confirmatory training using the full 800-epoch schedule and multiple random seeds

in order to validate the screening-stage findings and reduce variance in performance estimates.

2. We will extend the ablation study to include projection head variations and batch size–temperature interactions, which were identified in prior work as important factors for contrastive learning performance [1]

3. We plan to compare SimCLR with MoCo under matched architectures and augmentations to evaluate whether momentum-based contrastive learning improves robustness in small-batch regimes [2].

8 Code Availability

The implementation used in this project is based on an open-source SimCLR repository and extended to support the experimental ablations described in this paper.

The modified implementation and experiment scripts are available at:

- Project repository: `https://github.com/SAlaMusa/IDL`

- Base implementation: `https://github.com/sthalles/SimCLR`

9 Team Contributions

All team members contributed to literature review, experimental design discussions, and report writing and editing. The technical implementation work was divided as follows:

- **Marion Kipsang:** Implemented the SimCLR training pipeline and reproduced baseline experiments on CIFAR-10 and STL-10.

- **Sarah Moses:** Implemented single-augmentation ablation experiments and managed experiment configuration and training runs.

- **Millicent Malinga:** Implemented pairwise augmentation experiments to study interactions between transformations.

- **Philomena Githiri:** Implemented harmful augmentation experiments and generated evaluation tables and plots for result analysis.

References

[1] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton. A simple framework for contrastive learning of visual representations, 2020. URL `https://arxiv.org/abs/2002.05709`.

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